Coding challange 4

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2026-02-19

Contents

Link to my GitHub

https://github.com/Prasanna-cb/PLPA6820/tree/main/Coding%20Challange%204

Question 1

Explain the following

- a. The YMAL header is the top section an R Markdown file that tells R how to format and produce the output document.
- b. Literate programming is the approach that use the codes, explanations, and results in a single document so that easily understand to a reader about the the logic, purpose, and outcome of the analysis.

Question 2

Reference

- Noel, Z.A., Roze, L.V., Breunig, M., Trail, F. 2022. Endophytic fungi as promising biocontrol agent to protect wheat from Fusarium graminearum head blight. Plant Disease. https://doi.org/10.1094/PDIS-06-21-1253-RE

Loading necessary libries

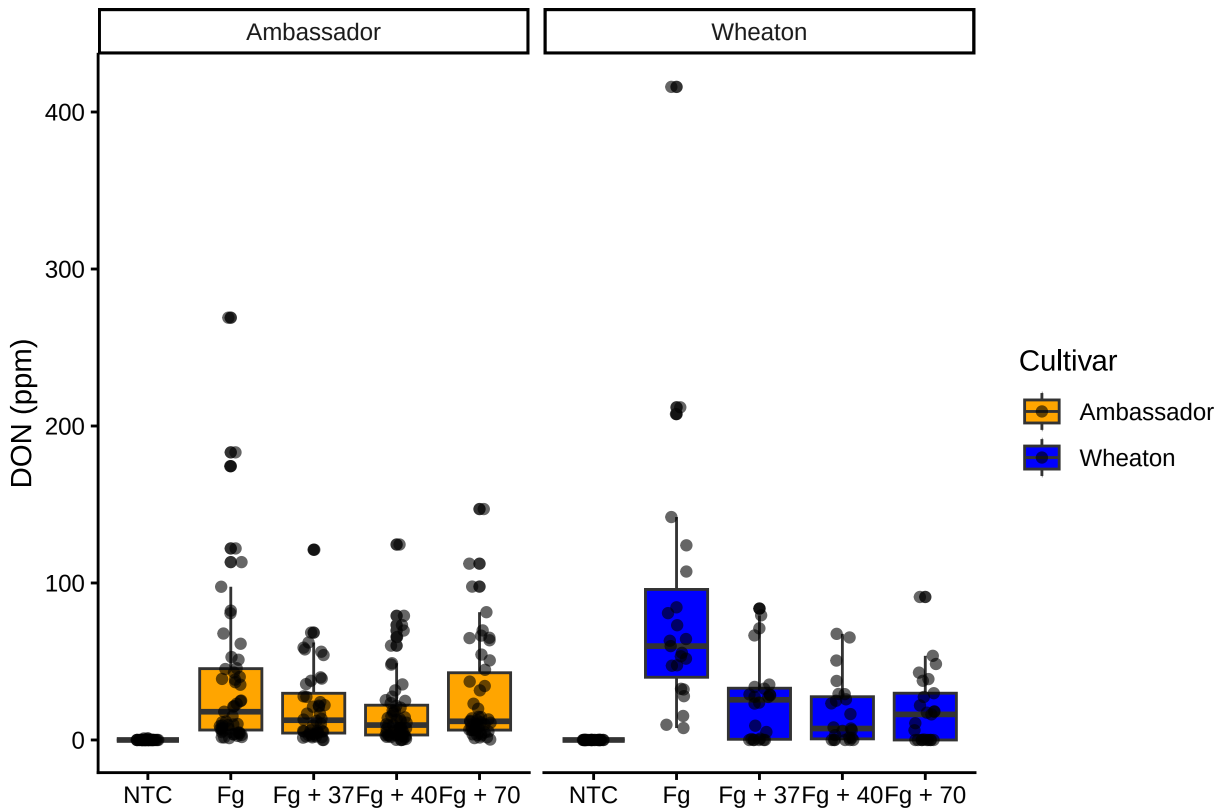
```
library(tidyverse)
library(ggpubr)
library(ggrepel)
```

Uploading the data to R

```
MycotoxinData=read.csv ("MycotoxinData.csv", na.strings = "na") #file path to GitHub repository
```

Createa a boxplot of DON by treatment and change the factor order level in X axis

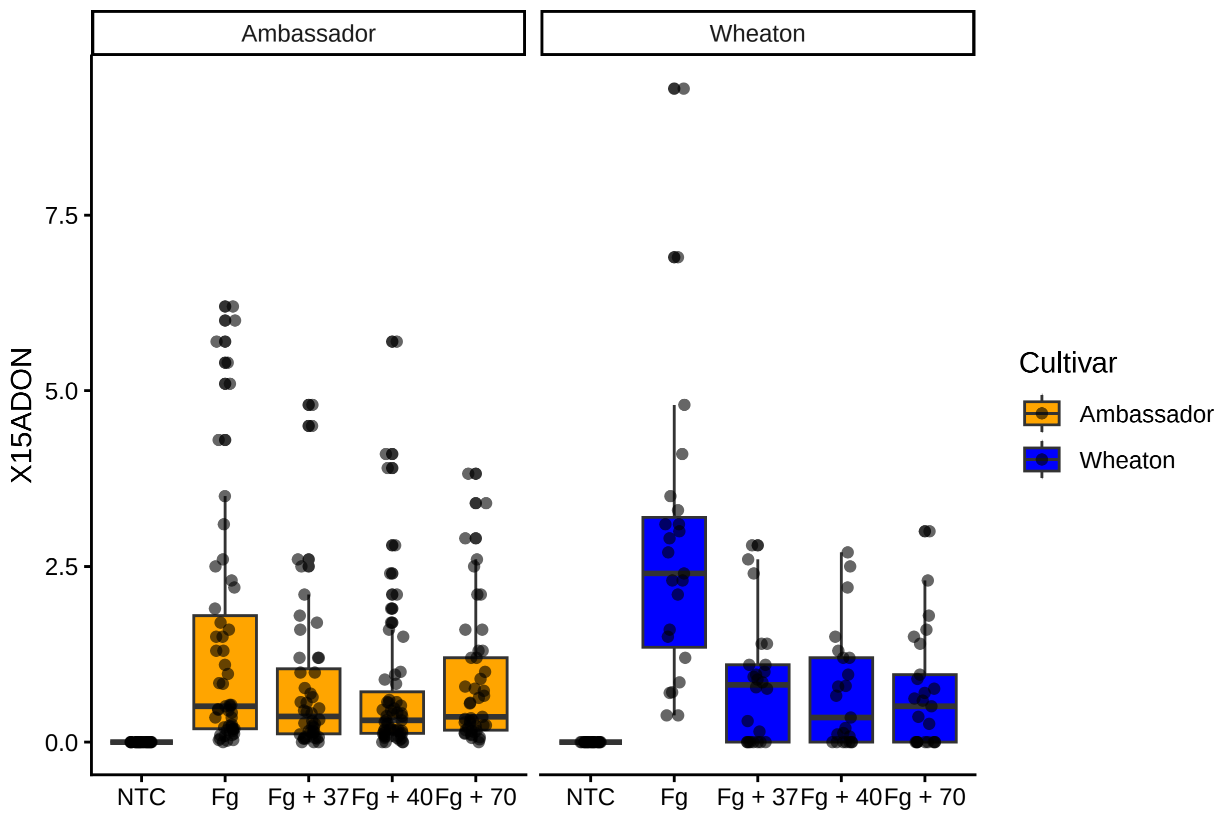
```
MycotoxinData$Cultivar <- as.factor(MycotoxinData$Cultivar)
    MycotoxinData$Treatment <- factor(MycotoxinData$Treatment,levels = c("NTC", "Fg", "Fg + 37", "Fg + 4
    Fig1 <- ggplot(MycotoxinData, aes(x = Treatment, y = DON, fill = Cultivar)) +
        geom_boxplot(position = position_dodge()) +
        geom_point(position = position_jitterdodge(), alpha = 0.6) +
        scale_fill_manual(values = c("orange", "blue")) +
        xlab("") +
        ylab("DON (ppm)") +
        facet_wrap(~ Cultivar) +
        theme_classic()
    print(Fig1)
```



Change the y-variable to plot X15ADON

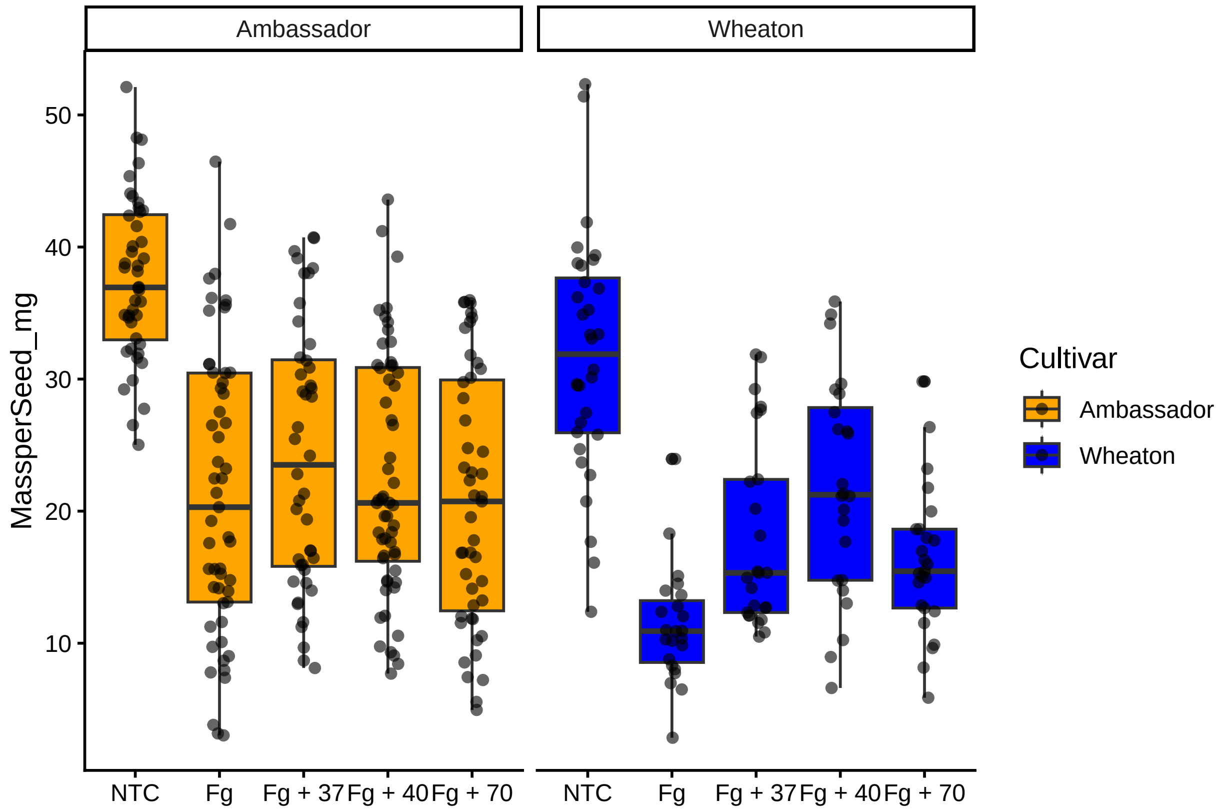
```
MycotoxinData$Cultivar <- as.factor(MycotoxinData$Cultivar)
     MycotoxinData$Treatment <- factor(MycotoxinData$Treatment,levels = c("NTC", "Fg", "Fg + 37", "Fg +
     Fig2 <-   ggplot(MycotoxinData, aes(x = Treatment, y = X15ADON, fill = Cultivar)) +
         geom_boxplot(position = position_dodge()) +
         geom_point(position = position_jitterdodge(), alpha = 0.6) +
         scale_fill_manual(values = c("orange", "blue")) +
         xlab("") +
```

```
        ylab("X15ADON") +
        facet_wrap(~ Cultivar) +
        theme_classic()
    print(Fig2)
```



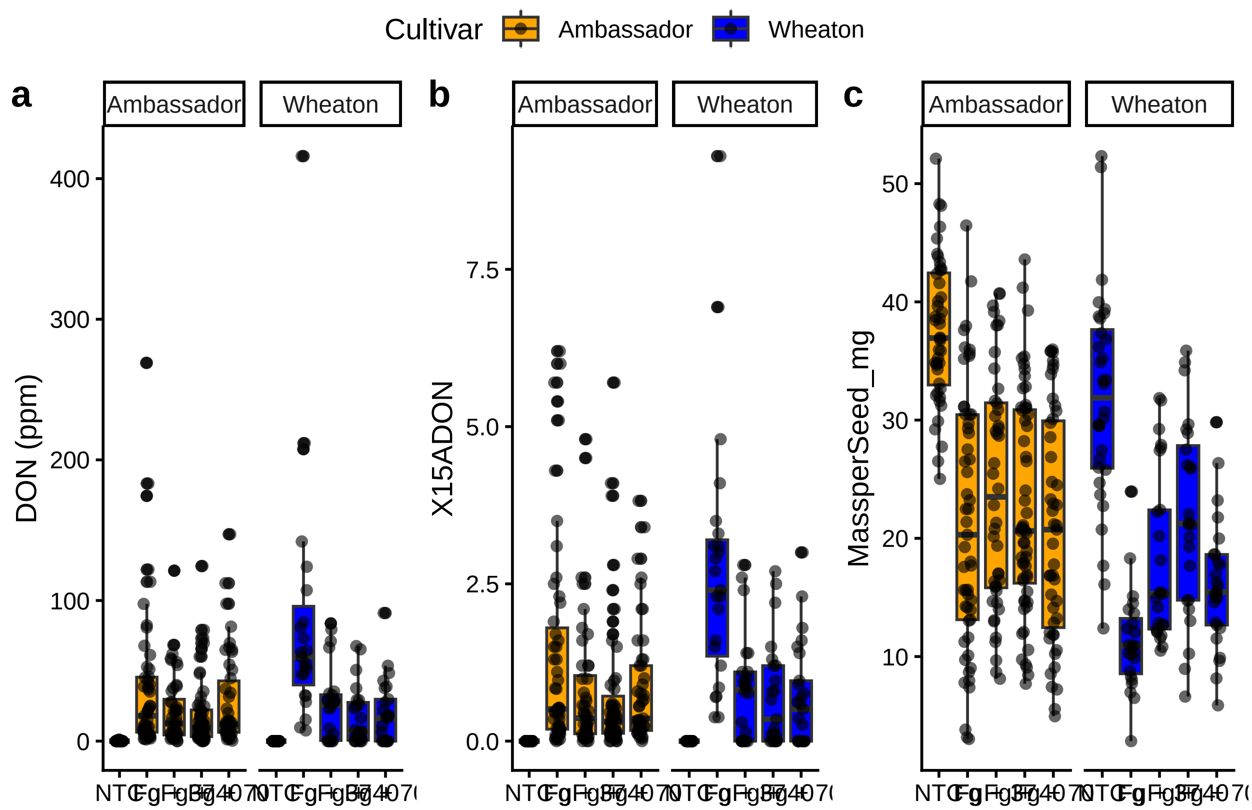
Change the y-variable to plot MassperSeed_mg

```
MycotoxinData$Cultivar <- as.factor(MycotoxinData$Cultivar)
    MycotoxinData$Treatment <- factor(MycotoxinData$Treatment,levels = c("NTC", "Fg", "Fg + 37", "Fg + 4
    Fig3 <-    ggplot(MycotoxinData, aes(x = Treatment, y = MassperSeed_mg, fill = Cultivar)) +
          geom_boxplot(position = position_dodge()) +
          geom_point(position = position_jitterdodge(), alpha = 0.6) +
          scale_fill_manual(values = c("orange", "blue")) +
          xlab("") +
          ylab("MassperSeed_mg") +
          facet_wrap(~ Cultivar) +
          theme_classic()
    print(Fig3)
```



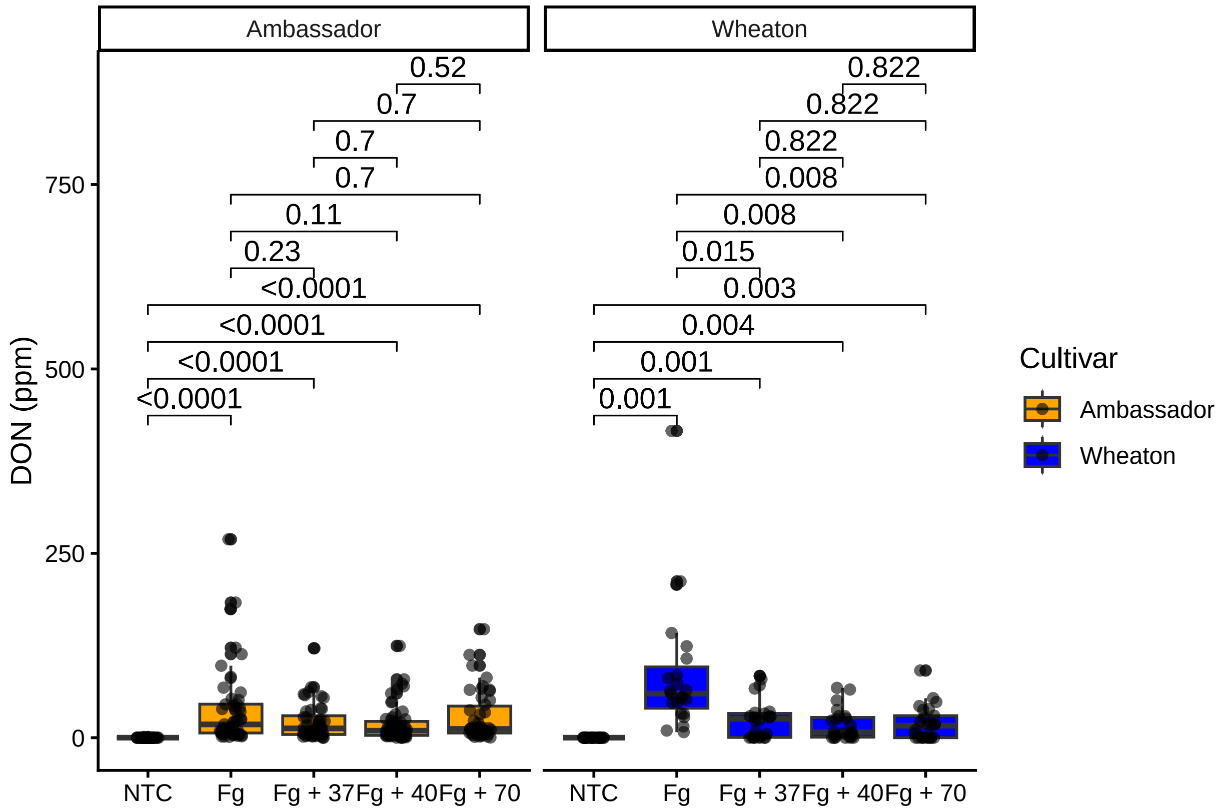
Use ggarrange function to combine all three figures

```
figure_combined <- ggarrange(
  Fig1,
  Fig2,
  Fig3,
  labels = "auto",
  nrow = 1,
  ncol = 3,
  common.legend = TRUE,
  legend = "top")
print(figure_combined)
```

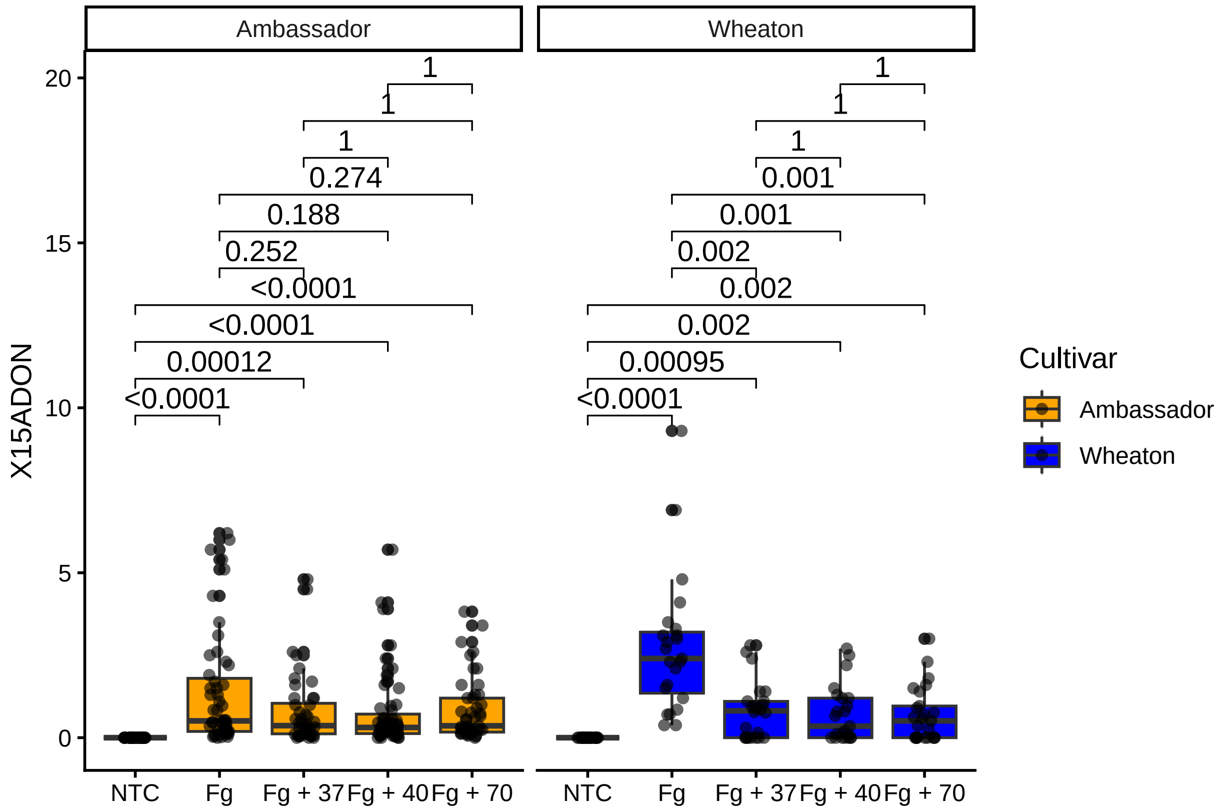


Use geom_pwc() to add t.test and pairwise comparisons to the above three plots

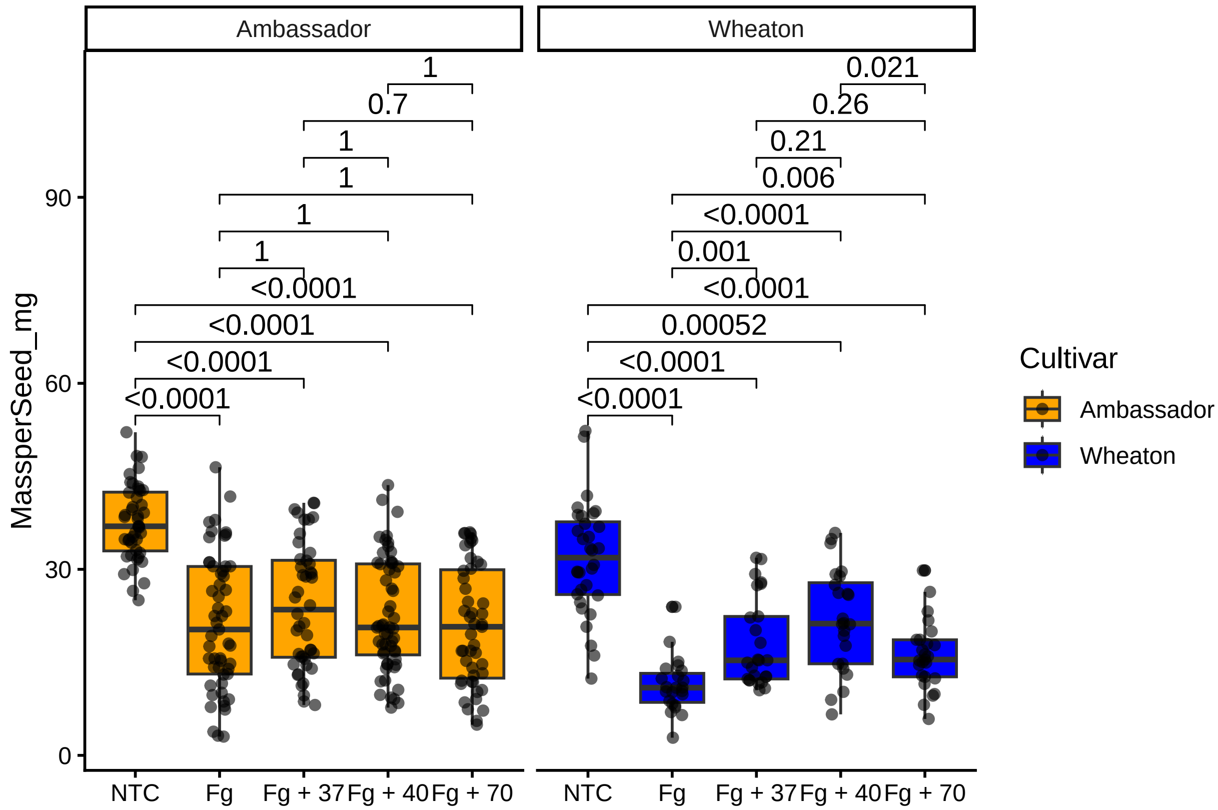
```
Fig1a <- Fig1 +
geom_pwc(aes(group = Treatment), method = "t_test", label = "p.adj.format")
print(Fig1a)
```




```
Fig2a <- Fig2 +
  geom_pwc(aes(group = Treatment), method = "t_test", label = "p.adj.format")
print(Fig2a)
```

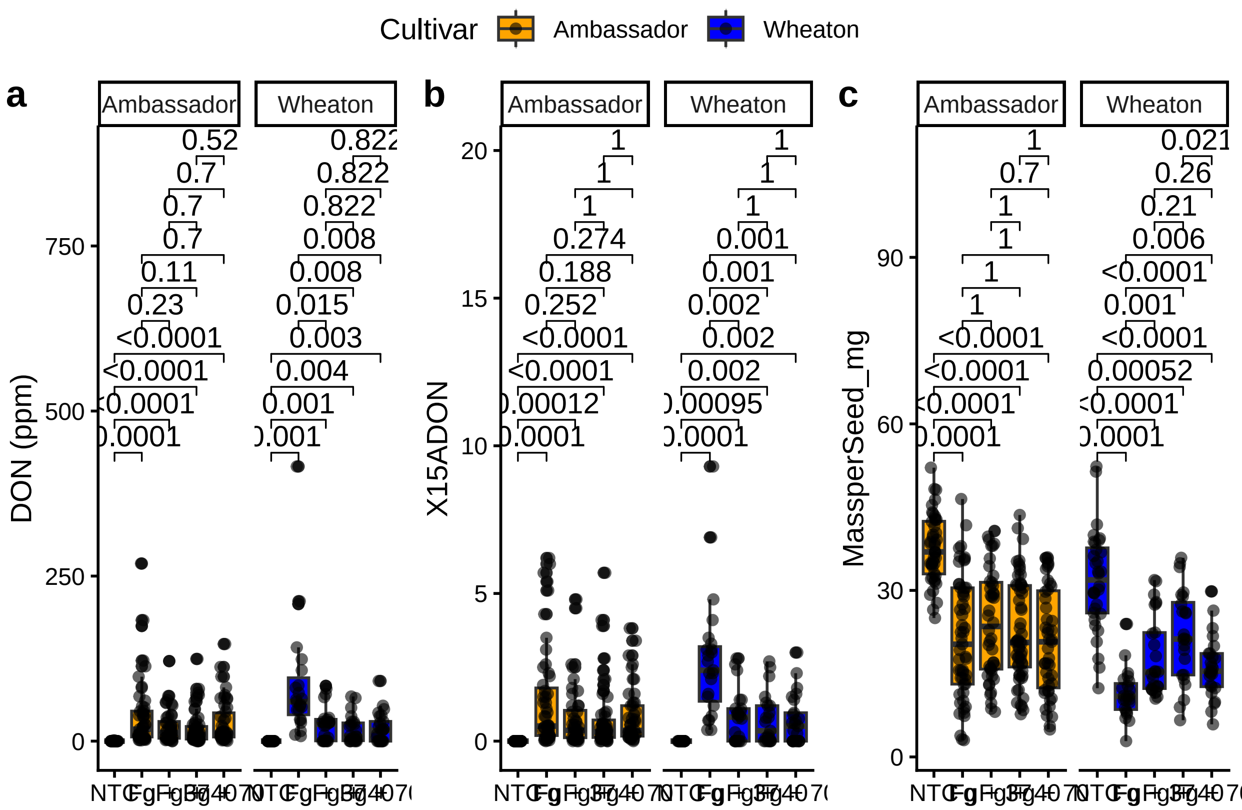



```
Fig3a <- Fig3 +
  geom_pwc(aes(group = Treatment), method = "t_test", label = "p.adj.format")
print(Fig3a)
```



Integration of pairwise comparison results into a single plot

```
figure_combined_final <- ggarrange(
  Fig1a,
  Fig2a,
  Fig3a,
  labels = "auto",
  nrow = 1,
  ncol = 3,
  common.legend = TRUE,
  legend = "top")
print(figure_combined_final)
```



Cultivar
Ambassador
Wheaton
a
b
c
DON (ppm)
X15ADON
MassperSeed_mg